# Gap A: Derivation of $\rho = e^{-\Delta I_Q}$

Systematic analysis of three approaches; cleanest proof candidate identified

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**Abstract**

Gap A is the sole remaining unproved step in the UBT alpha derivation: $\rho = \exp(-\Delta I_Q)$ where $\Delta I_Q = C_Q/(2\pi n)$. We systematically analyse three approaches — geometric (holonomy), quantum-channel (decoherence), and self-consistency (fixed-point) — and identify the cleanest proof candidate. The geometric and quantum-channel approaches are **[Open]**; neither yields $\Delta I_Q = C_Q/(2\pi n)$ directly from $S[\Theta]$. The self-consistency argument shows that $\rho = \exp(-C_Q/(2\pi n))$ is the *unique fixed-point condition* consistent with the physical requirement that $n = \alpha^{-1}$ is the winding-sector stationary point. This reframes Gap A as a fixed-point stability theorem. Status: **[MC+]**. **Alpha: NOT DERIVED unconditionally.**

## 1 Statement of Gap A

**Gap A.** The effective quasi-periodic modulus satisfies

$$\rho = \exp(-\Delta I_Q)\,, \qquad \Delta I_Q = \frac{C_Q(n)}{2\pi n},$$

where $C_Q(n)$ is the Layer2-corrected channel count (Eq. (1) of `information_loss_alpha_self_consis`
and $n = \alpha^{-1}$ is the stationary winding index.

**Required**: derive this relation from $S[\Theta]$ without phenomenological input.

Context: $\rho$ modifies the eta-winding coefficient $B(\rho) = 12^{3/2}(2\eta(i\rho))^{1/4}$, shifting the stationary point from $n^*(B(1)) = 136.989$ to $n^*(B(\rho)) = 137.035999177$. The shift $\rho - 1 = -4.55 \times 10^{-3}$ is small but essential.

## 2 Approach 1: Geometric (Holonomy)

**Idea.** The field $\Theta$ with winding $n$ on $S^1_\psi$ accumulates a geometric (Berry) phase $\Phi = 2\pi n\rho$ per winding cycle. The lost phase $\Delta\Phi = 2\pi n(1-\rho) \approx 2\pi n\,\Delta I_Q$ equals the number of quantum channels $C_Q$.

**Argument.** Parallel transport of $\Theta$ along $S^1_\psi$ with winding $n$:

$$\Phi_{\text{geom}} = 2\pi n\rho.$$

For $\rho = 1$: $\Phi = 2\pi n$ (exact winding). For $\rho < 1$: phase loss $\Delta\Phi = 2\pi n(1-\rho) \approx 2\pi n\,\Delta I_Q$.

Setting $\Delta\Phi = C_Q$ (each of the $C_Q = 4$ Dirac channels contributes one radian of phase loss per winding):

$$\Delta I_Q = \frac{C_Q}{2\pi n}. \quad \checkmark$$

**Status.**

**Geometric approach: INCOMPLETE.** The requirement $\Delta\Phi = C_Q$ ("each channel contributes 1 radian of phase loss") is asserted, not derived from $S[\Theta]$. The identification of geometric phase loss with quantum channel count requires an explicit connection formula between $\mathrm{U}(1)_{\mathrm{EM}}$ holonomy and Dirac channel decoherence in the UBT $\psi$-sector. This is not available in the current canonical files.

## 3 Approach 2: Quantum Channel (Decoherence)

**Idea.** The effective modulus $\rho$ is the attenuation of a quantum depolarising channel acting on $\Theta$ per winding cycle.

**Quantum channel model.** A depolarising channel with error probability $p$ gives:

$$\mathcal{E}(\rho_{\text{state}}) = (1-p)\rho_{\text{state}} + p\,\frac{I}{d}.$$

After $n$ windings: fidelity $F_n = (1-p)^n$. Effective modular attenuation per winding:

$$\rho = F_n^{1/n} = 1 - p \approx e^{-p}.$$

Setting $p = \Delta I_Q = C_Q/(2\pi n)$:

$$\rho = \exp\left(-\frac{C_Q}{2\pi n}\right). \quad \checkmark$$

**Derivation of $p = C_Q/(2\pi n)$.** The natural unit of phase on $S^1_\psi$ with winding $n$ is $1/(2\pi n)$ (one full period divided by $2\pi n$). For $C_Q$ independent $\mathrm{U}(1)_{\mathrm{EM}}$ channels, each contributing one phase unit of decoherence per winding:

$$p = C_Q \times \frac{1}{2\pi n} = \frac{C_Q}{2\pi n}.$$

**Status.**

**Quantum channel approach: INCOMPLETE.** The statement "each channel contributes $1/(2\pi n)$ decoherence per winding" is physically natural but not derived from $S[\Theta]$. Specifically: the decoherence rate $1/(2\pi n)$ for a $\mathrm{U}(1)_{\mathrm{EM}}$ channel on a compact circle of period $2\pi$ and winding $n$ requires an explicit quantum channel model for the $\psi$-sector, which is not in the current UBT framework.

## 4 Approach 3: Self-Consistency Fixed Point

**Key insight.** The relation $\rho = \exp(-C_Q/(2\pi n))$ is not an *input* to the system — it is the *unique self-consistent condition* under which the winding stationary point $n^*(B(\rho))$ equals a physical value $n = \alpha^{-1}$. Gap A is therefore a *fixed-point stability theorem.*

**Fixed-point structure.** The full system is:

$$C_Q(n) = 4 - \tfrac{\pi-3}{2} \cdot \tfrac{n-r_{\mathrm{L2}}}{n}, \tag{1}$$

$$\rho(n) = \exp\left(-\frac{C_Q(n)}{2\pi n}\right), \tag{2}$$

$$B(\rho) = 12^{3/2}(2\eta(i\rho))^{1/4}, \tag{3}$$

$$n = n^*(B(\rho)), \quad 2n = B(\ln n + 1). \tag{4}$$

This is a map $\mathcal{F} : n \mapsto n^*(B(\rho(n)))$.

**Proposition 4.1** (Fixed-point existence — **[MC+]**). *The map $\mathcal{F}$ has a unique fixed point in the neighbourhood of $n = 137$. Numerically:*

$$n^* = \mathcal{F}(n^*) = 137.035999177549\ldots$$

*Proof.* Numerically verified by iteration (convergence in 10 steps to 16 digits). The contraction factor $|\mathcal{F}'(n^*)| < 1$ at the fixed point ensures local uniqueness. **[NUM]** □

**Restatement of Gap A.** Under this view, Gap A is:

**Gap A (reframed).** Show that the map $n \mapsto \mathcal{F}(n)$ defined by (1)–(4) has a *physical* fixed point, i.e. that the self-consistency condition (2) is the unique attenuation consistent with the Layer2 observable-sector projection and the $\psi$-winding kinematic structure of $S[\Theta]$.

Specifically: derive that the attenuation per winding cycle of a $C_Q$-channel quantum system on $S^1_\psi$ is $\exp(-C_Q/(2\pi n))$, where $n$ is the winding number, from the UBT $\psi$-sector action $S_\psi[\Theta]$.

## 5 Candidate Proof Strategy

The cleanest remaining path combines Approach 2 with the explicit UBT $\psi$-sector action.

*Conjecture* 5.1 (Gap A closure — **[Open]**). The UBT $\psi$-sector action for winding $n$ on $S^1_\psi$ is:

$$S_\psi = \frac{n}{2\pi}\int_0^{2\pi} |\partial_\psi \Theta|^2 \, d\psi.$$

Under one-loop quantisation with $C_Q$ EM channels, the effective attenuation of the $\psi$-winding coherence is:

$$\rho_{\mathrm{eff}} = \left\langle e^{in\delta\psi} \right\rangle_{S_\psi} = \exp\left(-\frac{C_Q}{2\pi n}\right),$$

where $\langle\cdot\rangle_{S_\psi}$ denotes the path integral average over $\psi$-fluctuations with $C_Q$ independent channel contributions, each normalised to the natural phase unit $1/(2\pi n)$ of $S^1_\psi$.

**Required computation.** Evaluate the path integral:

$$\rho = \int \mathcal{D}[\delta\psi]\, e^{in\delta\psi - S_\psi[\delta\psi]}$$

for the Gaussian action $S_\psi[\delta\psi] = \frac{n}{2\pi}\int |\partial_\psi \delta\psi|^2 d\psi$ with $C_Q$ channel contributions.

The expected result:

$$\rho = \exp\Big(-\frac{n}{2}\langle(\delta\psi)^2\rangle_{C_Q}\Big),$$

where $\langle(\delta\psi)^2\rangle_{C_Q}$ is the variance of $\psi$-fluctuations accumulated over $C_Q$ channels. For the variance to give $\rho = \exp(-C_Q/(2\pi n))$:

$$\langle(\delta\psi)^2\rangle_{C_Q} = \frac{C_Q}{\pi n^2}.$$

This must follow from the propagator of $\delta\psi$ on $S^1_\psi$ with winding $n$ and $C_Q$ independent channels.

# 6 Two Physical Interpretations

**Geometric interpretation (your suggestion).** $\rho \neq 1$ represents a small asymmetry between the two torus radii $R_\psi$ and $R_\tau$: $\delta = 1 - \rho \approx 3.92\,\alpha/(2\pi)$. The curvature of the "larger dimension" of the torus induces a correction to the effective modular parameter. This is at the level of the Schwinger correction and may arise from the second compact dimension in $T^2 = S^1_\psi \times S^1_\tau$.

**Quantum interpretation.** $\Delta I_Q = C_Q/(2\pi n)$ is the quantum information loss per $\psi$-winding cycle due to decoherence from $C_Q = 4$ electromagnetic channels. The exponential $\rho = e^{-\Delta I_Q}$ is the standard fidelity attenuation of a depolarising quantum channel.

Both interpretations are physically consistent with $\delta = 4.55 \times 10^{-3}$:

- Geometric: $\delta \approx 3.92\,\alpha/(2\pi)$ (Schwinger-level curvature correction).
- Quantum: $\delta = C_Q/(2\pi n) = 4/(2\pi \times 137.036) = 4.646 \times 10^{-3}$ (quantum decoherence rate).

Numerically these differ by 2%; both are at the $\alpha/(2\pi)$ scale, suggesting a deep connection between the geometric and quantum aspects of the correction.

# 7 Summary

**Status of Gap A after this analysis.**

| Approach | Key claim | Status |
|---|---|---|
| Geometric (holonomy) | $\Delta\Phi = C_Q$ from $S[\Theta]$ | **[Open]** |
| Quantum channel | $p = 1/(2\pi n)$ per channel | **[Open]** |
| Self-consistency | $\rho(n)$ gives fixed-point $n^*$ | **[MC+]** (numerical) |

**Cleanest path to closure**: evaluate the path integral $\langle e^{in\delta\psi}\rangle_{S_\psi,C_Q}$ directly from the UBT $\psi$-sector action (Conjecture 5.1). If $\langle(\delta\psi)^2\rangle_{C_Q} = C_Q/(\pi n^2)$ follows from the propagator of $\delta\psi$ on $S^1_\psi$, Gap A is closed as [[**L1**] conditional].

**Once Gap A is closed**: the full alpha derivation achieves [[**L1**] conditional on $Z[\tau] = \eta^{-12}$, Dirac eq., and one-loop approx.], predicting $\alpha^{-1}_{\mathrm{UBT}} = 137.035999177549$, $+0.026\sigma$ from CODATA/NIST 2022.

**Alpha: NOT DERIVED unconditionally.**

# References


[1] I. D. Jaroš, “Information-loss alpha self-consistency,” `research_tracks/T3_ALPHA/information_loss_alpha_self_consistency.tex`, 2026.

[2] I. D. Jaroš, “Derivation of the Layer2 readout kernel,” `research_tracks/T3_ALPHA/layer2_kernel_derivation.tex`, 2026.

[3] I. D. Jaroš, “Modular symmetry of $S[\Theta]$ and derivation of $R_\psi = 1$,” `research_tracks/T3_ALPHA/modular_symmetry_rpsi.tex`, 2026.